THE LANGUAGE OF BUSINESS ANALYTICS

1) Foundational Concepts of Analytics

1) Business Analytics

Studying a business to help it work better and make better decisions. It helps implement and evaluate recommended solutions.

Real world example

Retail store notices a drop in sales.
Business analyst, analyzes sales data and customer feedback. Then make recommendations based on data received.

Source
ibm.com/think/topics/business-analytics.

2) Data Analytics

The process of turning raw data into meaningful information. Basically it is the collecting, cleaning and analyzing data to make useful insights. There's different tools to be used such as excel, Python and SQL, PowerBI.

Real world example.

An online store analyzes sales data.
- Finds that electronics sells more than the other items.
- The online store decides to stock more electronics to have increased sales and have increased shortagts.

Source.
coursera.org/articles/data-analytics.

3) Business Intelligence (BI)

Is the process of collecting analyzing and presenting business data to help organizations make informed decisions. Provides accurate real-time information for smarter decisions.

Real world example.

A company uses BI dashboard to track sales.
- Sees that one product is selling more in a specific region.
- Decides to increase stock and marketing in that area, resulting in increased revenue

Source 8
- ibm.com/think/topics/business Intelligence.
- Chat GPT

04) Data-Driven Decision Making.

- It is the process of making decisions based on collected data, analysis and evidence rather than guesswork. Helps businesses avoid costly mistakes by relying on facts.

Real word example

A clothing store tracks sales data:

- Notices that dresses sell more in summer.
- Decides to stock more dresses before summer.

Resulting in increased sales

Source:

- Chat GPT

5) Big Data

Refers to massive, complex data sets that traditional data management systems cannot handle.

* Volume - Data is more because there is more of it.
* Velocity - Is the speed at which data flows into a system and big data moves quickly.
* Variety - refers to the many different formats that big data can take.

e.g Structured, unstructured and semi-structured.

Real world example

A social media platform collects Posts, likes and comments, videos.

- Millions of users generating data every second.

Source:

ibm.com/think/topics/big-data.

06) Descriptive Analytics

- Summarizing past data to see what happened. It describes trends and relationships. It works by aggregating, cleaning and visualizing data to create metrics and KPIs for dashboards

Real world example

A hospital might track average emergency room wait times, most common diagnoses and and yearly admission trends to identify operational improvements.

Souce:

- Chat GPT.

07 Diagnostic Analytics

Is the process of analyzing historical data to uncover the root causes of trends. It goes beyond descreptive analytics, which explains what happened, by answering why it happened. By identifying the reasons behind outcomes, it helps make data-driven decision-making possible for organizations.

Real world example.

An online store experienced a drop in sales and used diagnostic analytics to invertige the cause.

Source:

ibm.com/think/topics/diagnostic-analytics
Chat GPT.

08 Predictive Analytics

Is a data-driven approach that uses historical data, statistical algorithms and machine learning techniques to predict future outcomes. It identifies patterns and trends in past data to predict future events enabling organizations to make informed decisions.

Real world example

A retail store wants to know future sales trends.
- Analyze past sales data
- Look at seasonal trends.
- Use statistical models to predict future demand.

Source:

Chat GPT

09 Prescriptive Analytics

Is the practice of analyzing data to identify patterns which can be used to make predictions. It is a practice situated within the disciplines of business analytics and business intelligence.

While all 4 types of analytics are usefull to tell the story within data, prescriptive analytics differs from the other types in its focus on not only predicting future outcomes but also recommending actions.

(10) Data Literacy

The ability to read, understand, analyze, and communicate with data to make informed decision.

It consists of 4 core Abilities.

- Read data
- Work with data
- Analyze data
- Argue with data.

2 DATA, MEASUREMENT & REPORTING

(11) Structured Data.

- Data comes in a structured or consistent format. Stored in table with rows and columns

Real world example

Bank transactions and Student records

(12) Unstructured Data

- Data sets that do not follow a structure. Doesnt follow a particular format.

Real world example

- Emails and text documents.

(13) Data Warehouse

- Is a central place where we store multiple data-databases.
- It aggregates data from various sources into a central data store for queing and analysis.

(14) Data Mining

Process of analyzing large sets of data to discover hidden patterns, trends and useful insights. It helps businesses make informed decisions and improve marketing strategies

(15) Data Visualization

The process of presenting data in a visual format - such as graphs, charts or dash boards, so that it is easier to understand and intepret. It also reveals patterns and trends which supports decision making

Real world example

Company creates dashboard showing monthly production using line graph.

Managers can quickly see which month performed badly

Source:

Chat GPT.

(16) Dashboard

- A visual display of important information, typically shown using charts, graphs and tables, all one place to help monitor performance and make decisions.
- It tracks perfomance as data updates

Real world example:

A sales dashboard shows
- Total sales for the month
- Sales by region.
- Top selling products

A manager can see quickly;
- Which region is performing best.
- Identify products that need improvement.

Source:
Chat GPT.

(17) Metric

- A number that shows how well something is performing.
- Any measurement that provides information using data

Example
Online Store website
* 15000 website visitors
* 500 item bought.

(18) Key Performance Indicator (KPI)

It is a specific metric tied to a goal, like increasing monthly visitors from 10 000 to 15 000.
All KPIs are metrics but not all metrics qualify as KPIs.
- To select good KPIs, you must identify your goal, determine which metrics best track progress toward it.

Source:
What are KPIs and Metrics; YouTube by Alex the Analyst.

(19) Benchmark

Is a standard or reference point used to compare performance and evaluate how well something is doing.

(20) Segmentation.

Is the process of dividing a large group of data/customers into smaller, similar groups based on shared characteristics.

03 CUSTOMER AND SALES KPIs

21 Conversion Rate

Conversion rate is the percentage of people who take a desired action out of the total number of people who had the opportunity to take that action.

In simple terms:

Conversion rate = $\frac{\text{Number of conversion}}{\text{Total visitors}}$

22 Customer Acquisition Cost (CAC)

Is the cost related to acquiring a new customer

- It refers to the resources and costs incurred to acquire an additional customer

23 Customer Lifetime Value (CLV/LTV)

Is the total worth of profit from a customer to a business over the entrely of their relationship.

- It is one of the most important metrics for tracking customer experience and value.
- CLV looks at how valuable a customer is to the organization as a whole.

Source:

Corporate finance institute.com/resources/accounting

24 Churn Rate

Is the percentage of customers or subscribers who stop doing business with a company over a specific period.

- It is used to measure the percentage of subscribes or employees who leave a company during a specific period.

25 Customer Retention Rate

- Measures the percentage of customers who continue to use a company's product or service over a specific period.

Source:

Investopedia.com/terms/churn rate.asp

(26) Net promoter Score (NPS)

- Is a standardized customer loyalty metric that measures the willingness of customers to recommend a company's products or services to others
- It is measured through a simple NPS survey that asks one question: "How likely are you to recommend this company to a friend or colleague?"

(27) Average Order Value. (AOV)

- Measures the average amount a customer spends per transaction.
- It is a key metric for optimizing revenue and marketing efficiency.
- Provides some insights regarding customer behavior, for example:
  a lower AOV usually indicates that the seller's customers prefer to make small purchases with each order.

Source:
corporatefinanceinstitute.com/resources/valuation/average-order-valu

(28) Sales Funnel

- Is a marketing concept that describes the journey a potential customer takes from the first interaction with a business to the final purchase decision.
- Sellers use this funnel to track each prospects progress toward close.

(04) MARKETING AND CAMPAIGN ANALYTICS

(29) Marketing Campaign

- Is a planned effort by a business or organization to communicate a message about a product, service or idea.
- This is done using various forms of media.

(30) Post-Campaign Analysis

- Evaluates a marketing campaign's performance to identify successes and areas of improvement.
- It involves collecting data measuring key performance indicators (KPIs) and understanding what worked and what didn't.

(31) Incremental Revenue.

- It is the additional revenue generated from an increase in sales such as a marketing campaign.
- It focuses on the difference between revenue with the action taken and revenue without it.

Source:
accountingtools.com/articles/incremental-revenue.

(32) Return on Investment (ROI).

- Is a financial metric that measures the profitability or efficiency of an investment relative to its cost.
- Its a ratio used to evaluate the gain or loss generated by an investment compared to its initial cost.

(33) Return on Ad Spend (ROAS)

- Measures the revenue generated for every dollar spent on advertising.
- It is a key marketing metric that evaluates the effectiveness of advertising campaigns.

$$ROAS = \frac{\text{Revenue Generated}}{\text{Ad Spend}}.$$

(34) Click-Through Rate (CTR)

- The percentage of users who click on a link after seeing it, serving as a key metric for measuring engagement and ad effectiveness.
- A higher CTR indicates an ad has effectively captured attention.
- CTR is crucial for evaluating the effectiveness of online marketing campaigns.

Source:
Invertopedia.com/terms/click/click-throughrates.asp

(35) Bounce Rate

- Is the percentage of website sessions in which visitors leave after viewing only one page without interacting further.

(36) A/B Testing

Is a method of comparing two or more versions of a product to determine which one performs better.
- Helping to make data-driven decisions.

(37) Lead Generation

- Is the process of attracting potential customers, capturing their interest.
- Involves identifying and cultivating potential customers for a business's products.

(38) Marketing Attribution

The process of measuring and assigning credit to marketing channels that influence a company's revenue and pipeline.
- Its the means by which the customer came to know and buy your product or service.